

SAN JUAN LA PUNTILLA, Puerto Rico

WMO: 994043

Lat: 18.459N

Lon: 66.116W

Elev: 16

StdP: 14.69

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	72.0	73.0	N/A	N/A	N/A	N/A	N/A	N/A	32.8	79.0	27.2	78.6	5.5	220	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	7.3	88.3	N/A	87.3	N/A	86.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11.1	140

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.4	21.0	19.1	70.3	91.4	0.9	1.5	69.6	92.4	69.1	93.3	68.6	94.1	67.9	95.2		
			DB	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.3	77.4	77.4	77.4	78.8	80.5	82.2	82.2	82.7	82.9	82.5	80.6	78.7
	DBStd	2.58	1.29	1.37	1.53	1.81	1.77	1.39	1.32	1.29	1.37	1.40	1.42	1.54
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11060	851	768	851	863	946	965	999	1014	987	1008	918	890
	CDD65	5585	386	348	386	413	481	515	534	549	537	543	468	425
	CDH74	56701	2831	2504	2741	3575	4817	5956	6301	6637	6474	6309	4836	3720
	CDH80	13602	133	127	149	379	922	1784	1977	2278	2281	2048	1102	422
Wind	WSAvg	9.8	9.9	10.2	10.0	10.8	9.5	10.4	11.7	10.9	8.2	8.0	8.1	9.5
Precipitation	PrecAvg	68.3	4.1	2.3	3.0	5.1	6.1	5.7	5.9	7.8	7.3	7.5	7.3	6.5
	PrecMax	103.3	9.0	4.6	8.9	15.2	17.0	18.4	10.1	14.4	12.5	17.6	17.3	12.0
	PrecMin	42.8	0.6	1.1	0.6	1.0	1.0	0.7	2.9	3.1	2.6	4.0	2.6	0.4
	PrecStd	13.1	2.0	0.9	2.0	3.6	4.1	3.6	2.1	3.1	3.2	4.4	4.5	3.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.0	84.9	85.2	87.9	88.0	88.3	87.8	89.1	89.7	89.1	87.0	85.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	82.2	82.8	82.7	85.1	86.3	87.1	86.4	87.1	88.1	87.7	85.5	83.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	81.3	81.5	81.5	83.4	85.1	86.1	85.6	86.1	87.0	86.7	84.8	82.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	80.7	80.4	80.6	81.9	83.8	85.2	85.0	85.4	86.2	85.9	84.2	82.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.8	6.4	6.4	6.3	6.2	5.9	5.7	6.0	7.3	7.3	7.2	6.7
		MCDBR	7.6	8.0	7.6	8.3	7.8	7.1	6.3	6.5	8.2	8.0	7.9	7.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.371	0.386	0.386	0.413	0.453	0.527	0.525	0.488	0.442	0.391	0.385	0.377	
	taud	2.551	2.495	2.492	2.444	2.345	2.098	2.102	2.235	2.394	2.553	2.569	2.563	
	Ebn at Noon	290	290	293	284	269	247	248	259	272	285	283	284	
	Edh at Noon	31	34	35	37	41	51	51	45	38	32	30	30	
All-Sky Solar Radiation	RadAvg	1428	1663	1840	1933	1860	1899	1922	1873	1768	1587	1392	1338	
	RadStd	51	36	77	109	129	109	96	60	114	102	102	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.50	+0.85	N/A	+0.59	N/A	N/A	N/A	N/A	+203	+202

Nomenclature: See separate page